

PHYSICS, KINETICS, AND SHIELDING OF FAST BREEDER REACTORS—SURVEY OF RECENT WORK DONE IN FRANCE

F. STORRER,* M. CADILHAC,† J. M. CHAUMONT,* P. P. CLAUZON,* F. EBER-
SOLDT,* I. PUIG,* P. GOVAERTS,* A. KHAIRALLAH,* T. LACAPELLE,
G. NEVIERE* and J. RASTOIN‡

Work done under the auspices of the Euratom-CEA Association for
fast neutrons

1. Introduction

The physics characteristics of large dilute fast reactors cannot as yet be determined very accurately, because of insufficient basic data and of inadequate theoretical methods. A correct interpretation of critical and other integral experiments and the application of their results to the calculation of power reactors also depend heavily on adequate theoretical methods; this is particularly so when not enough fuel is available to perform full scale critical mockups and when flexible criticals are used which can never simulate exactly the composition and the lattice structure of power reactors. This review paper of some work done in France in recent years covers, except in the case of shielding, only the theoretical aspects. Considerable work is also being done on basic data measurements and evaluations, and a very important integral-experiment programme is under way, which will soon begin to yield results. The theoretical topics of current interest which are discussed here appear in the section headings.

The two paramount problems which still await a satisfactory treatment are the elastic slowing down at low energies and the variation of the spectrum within the usual coarse groups in the vicinity of interfaces. The first problem is not adequately treated since experimental results indicate that calculations always underestimate the low energy tail of the spectrum. A good knowledge

* CEN, Cadarache.

† University of Marseille.

‡ CEN, Fontenay-aux-Roses.

of this part of the spectrum and of the adjoint spectrum is essential for an accurate determination of reactivity coefficients of various materials and of the Doppler effect. Two aspects of this problem are touched in sections 3 and 8.

The second problem, which is heavily dependent on the scattering resonances is equally important for the calculation of power reactors and for the analysis of zoned critical experiments, as discussed in section 11. This problem is not easy to solve, since it involves at the same time a fine spatial and energetic resolution. No economic way to treat it has yet been developed (see section 9).

2. Influence of the Heterogeneity of Fast Reactors on the Doppler and Sodium-loss Coefficients

Doppler and sodium-loss coefficients in fast reactors are usually calculated under the assumption that the reactor is homogeneous. A few authors have used approximate or numerical methods to account for heterogeneity.^(1, 2) A method is presented here for investigating the influence of heterogeneity on these coefficients with essentially no more assumptions than for the homogeneous calculations.

In a previous report,⁽³⁾ a perturbation method was presented for calculating the effect of heterogeneity on k_{eff} and on the spatial fine structure of the flux without accounting for the energy fine structure of the cross-sections. The core was replaced by an equivalent infinite medium (by the use of fictitious capture cross-sections representing the leakage) that was divided into a certain number of homogeneous regions, generally two: the fuel region and the diluent region. The collision probability formalism was used in the integral transport operator, and the effects of heterogeneity were calculated by a perturbation method, the homogeneous core being the unperturbed reference and the perturbation being due to the change in the collision probabilities when going from the homogeneous core to the heterogeneous one. Numerical results, obtained with the code PERHET, were presented.

The investigation of the heterogeneous Doppler and sodium-loss coefficients⁽⁴⁾ which is summarized here is based on an extension of the method of ref. (3). The model used is that of the infinite equivalent core, and the formalism is the same, except for two main differences: (a) the reference core is the actual heterogeneous one (before the change in the fuel temperature or the sodium loss) the k_{eff} , flux and adjoint flux of which are calculated by the method of ref. (3); (b) the perturbation formulae for the Doppler and sodium-loss coefficients contain not only terms proportional to the change in the collision probabilities but also terms proportional to the change in the cross-sections of either the fuel region (Doppler) or the diluent region (sodium-loss).

Namely, the general perturbation formula, applicable to both coefficients is:

$$\frac{\delta k_{\text{eff}}}{k_{\text{eff}}} = \frac{\sum_{g=1}^N \sum_{j=1}^n \phi_j^g \left\{ -\delta \mu_{ij}^g \phi_j^{*g} + \sum_{k=1}^N \sum_{i=1}^n (\delta \mu_j^{g \rightarrow k} P_{ji}^k + \delta P_{ji}^k \mu_j^{g \rightarrow k}) \phi_i^{*k} \right\}}{\frac{1}{k_{\text{eff}}} \sum_{g=1}^N \sum_{j=1}^n \phi_j^g \nu \mu_f^g \sum_{k=1}^N \sum_{i=1}^n \chi_x^k P_{ji}^k \phi_i^{*k}} \quad (1)$$

where ϕ_j^g and ϕ_j^{*g} are the flux and adjoint flux⁽³⁾ in energy-group g and region j of the heterogeneous core. P_{ij}^g is the collision probability in region j of group g neutrons born in region i . μ_{ii}^g , μ_{fj}^g and $\mu_j^{g \rightarrow k}$ are the total, fission and energy-transfer to group k (by scattering and fission) cross-sections for neutrons of group g in region j . δP_{ij} and $\delta \mu$ are the changes in P_{ij} and μ , due to Doppler broadening or sodium loss.

The cross-sections are macroscopic cross-sections per unit volume of the whole core.

The physical meaning of equation (1) is clear enough. For the subsequent discussion, however, it will be convenient to introduce a new quantity $\bar{\phi}^*$ defined by

$$\bar{\phi}_j^{*g} = \sum_{i=1}^n P_{ji}^g \phi_i^{*g}$$

which has the same meaning as the conventional importance function, and to write equation (1) in the following form:

$$\frac{\delta k_{\text{eff}}}{k_{\text{eff}}} = \frac{1}{I_f} \left\{ - \sum_{g,j} \phi_j^g \delta \mu_{ij}^g \bar{\phi}_j^{*g} + \sum_{g,j} \phi_j^g \sum_{k=1}^N \delta \mu_j^{g \rightarrow k} \bar{\phi}_j^{*k} + \sum_{g,j} \phi_j^g \left[\sum_{k,i} \delta P_{ji}^k \mu_j^{g \rightarrow k} \phi_i^{*k} + \delta \mu_{ij}^g \sum_{i=1}^n P_{ji}^g (\phi_i^{*g} - \phi_j^{*g}) \right] \right\} \quad (2)$$

where I_f represents the denominator of equation (1), i.e. the so-called importance of fission neutrons.

Equation (2) can be applied in particular to the homogeneous core, by putting the appropriate values of the δP_{ij} 's and putting ψ and ψ^* for ϕ_j and $\bar{\phi}^*$, respectively, ψ and ψ^* being the classical flux and adjoint flux in the homogeneous core.

Conventional first order perturbation theory would give, for the homogeneous core, the following equation:

$$\frac{\delta k_{\text{eff}}}{k_{\text{eff}}} = \frac{1}{I_f} \left\{ - \sum_{g,j} \psi_j^g \delta \mu_{ij}^g \psi_j^{*g} + \sum_{g,j} \psi_j^g \sum_{k=1}^N \delta \mu_j^{g \rightarrow k} \psi_j^{*k} \right\} \quad (3)$$

and it can be shown that equations (2) applied to a homogeneous system and (3) do not generally give the same result, except if the relative change in the cross-sections is infinitely small. This is because the third term in equation (2),

which corresponds to the spatial shift of neutrons, does not vanish, but is generally positive for both Doppler and sodium loss effects. Its relative importance will be discussed later for each of these effects.

Partial Sodium-loss Effect

The usual assumption is made that the influence of the sodium loss on the effective cross-sections of the resonant fuel (change of the energy self-shielding) is negligible. The change in reactivity is given by equation (2) where the cross-sections and collision probabilities perturbations are due only to sodium removal. It is clearly seen that the three terms correspond respectively to the capture, spectral shift, and spatial shift components. The leakage component is not accounted for here, and will be discussed below. It should be noted that the spatial shift component, which is not taken into account in classical perturbation theory, is not zero even in the homogeneous case. This last component is, however, of second order in the cross-section perturbations.

Numerical Results and Discussion

The reference core contains about 35 vol.% UO_2 - PuO_2 fuel (isotopic ratio about 5), 14 vol.% SS and 51 vol.% Na. Two heterogeneous versions of this core were considered, a power reactor (oxide fuel, $\phi = 6$ mm) and its critical mockup (metal fuel, $\phi = 12$ mm, oxygen mixed with the diluent). The

TABLE 1

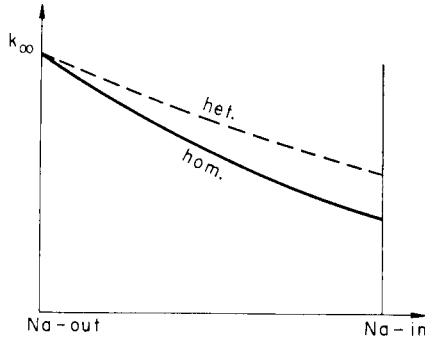
Component	Homogeneous classical	Homogeneous this method	Power reactor this method	Critical mockup this method
Capture	$+0.168 \times 10^{-3}$	$+0.168 \times 10^{-3}$	$+0.164 \times 10^{-3}$	$+0.160 \times 10^{-3}$
Spectral and spatial shift	$+0.546 \times 10^{-3}$	$+0.775 \times 10^{-3}$	$+0.755 \times 10^{-3}$	$+0.620 \times 10^{-3}$
Total	$+0.714 \times 10^{-3}$	$+0.940 \times 10^{-3}$	$+0.920 \times 10^{-3}$	$+0.780 \times 10^{-3}$

ABBN⁽⁴⁾ cross-section set was used, with resonance self-shielding corrections corresponding to the homogeneous core.

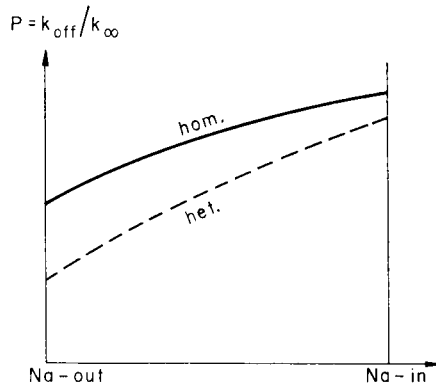
Table 1 summarizes the results obtained for a 10% sodium loss. It is seen that heterogeneity influences appreciably the total positive component of the sodium loss effect. Two main conclusions appear from Table 1:

- (a) The method of this paper, when applied to a homogeneous core, gives for a finite sodium removal a more positive Na loss effect than that given by conventional perturbation theory. This is because the second

order spatial shift term in equation (2) is not negligible, even for a sodium loss as small as 10%. This also demonstrates the effectiveness of this method, since it is well known that conventional perturbation theory largely underestimates the Na loss effect.



- (b) The heterogeneity tends to decrease the positive Na loss effect. This is well illustrated by the previous diagram. The full curve corresponds to the homogeneous case, with a positive sodium coefficient. The curve is concave upwards because of sodium self-shielding. The dashed line corresponds to the heterogeneous case. For Na-in the heterogeneous case is more reactive, as is usually the case for fast systems. For Na-out, however, the homogeneous and heterogeneous cases are equivalent since empty channels have no effect in an infinite medium.



From the assumptions underlying the first diagram it follows directly that the positive sodium component must be smaller in the heterogeneous case than in the homogeneous one, as is confirmed by our calculations.

The reactivity increase resulting from total sodium loss is reduced exactly by the amount of reactivity gained in the Na-in case by heterogeneity.

The (negative) leakage component of the sodium coefficient is illustrated by the second diagram, with the same abscissa as the preceding one, but with the non-leakage probability $P = k_{\text{eff}}/k_{\infty}$ as ordinate. The heterogeneity reduces P , because of the streaming effect, which is certainly more important in the Na-out case. Lacapelle⁽⁶⁾ studied this effect by a method based on the work of Benoist⁽⁷⁾ and found that heterogeneity increases the negative leakage component by a few per cent.

Conclusions

Thus the net sodium coefficient is certainly less positive in the heterogeneous than in the homogeneous case. Heterogeneity may even turn a positive coefficient into a negative one, particularly in view of the fact that the net sodium coefficient, being a small difference of two large components, is very sensitive to corrections on these components. Further work is in progress in order to analyse these phenomena quantitatively for a variety of media.

As indicated in ref. (3), the perturbation method presented in this paper can be carried to higher orders, and thus give accurate results even for total sodium loss: the flat-source approximation which is used in this method becomes certainly more valid when sodium is removed. However, when wide energy groups are used, one set of cross-sections cannot correctly describe the elastic slowing down for different sodium concentrations, because of the changing neutron spectra within the groups. Therefore the possibility of extending this method to several hundred groups, so as to resolve the scattering resonances, is at present being examined.

Doppler Coefficient

Spatially constant temperatures and effective cross-sections are assumed inside the fuel elements; the diluent cross-sections are independent of temperature. In these conditions, the main problem in the calculation of the heterogeneous isothermal Doppler effect remains to find the proper changes in the group effective cross-sections and group collision probabilities to be used in equation (2). The method used is reported elsewhere⁽⁴⁾ where the validity of the various approximations is also discussed. Only numerical results are presented here.

The temperature and dilution dependent Fischer's⁽⁸⁾ reaction cross-sections were interpolated and used for the Doppler calculations. These were performed for the same cases for which the sodium loss effect was calculated.

Table 2 compares the values of $T(\delta k_{\text{eff}}/\delta T)$ Doppler for the homogeneous core and the power reactor at 1200°K, and for the homogeneous core and the critical mockup at 300°K. The column labelled "exact" refers to the method presented here, while the label "approximate" refers to the method of

TABLE 2

Contributions	Homogeneous 1200°K		Power reactor 1200°K		Homogeneous 300°K		Critical mockup 300°K	
	Exact		Exact		Exact		Exact	
U-238	-0.565×10^{-2}		-0.570×10^{-2}		-0.545×10^{-2}		-0.597×10^{-2}	
Pu-239	$+0.111 \times 10^{-2}$		$+0.113 \times 10^{-2}$		$+0.137 \times 10^{-2}$		$+0.179 \times 10^{-2}$	
Total	-0.454×10^{-2}		-0.457×10^{-2}		-0.408×10^{-2}		-0.418×10^{-2}	
							Approx.	
							-0.608×10^{-2}	
							$+0.181 \times 10^{-2}$	
							-0.427×10^{-2}	

ref. (1), which uses the homogeneous perturbation equation but with the heterogeneous changes in the effective cross-sections.

Some interesting conclusions may be drawn from the results of Table 2:

- (a) heterogeneity seems to have a negligible effect on the total Doppler effect, as well for the metal fuel as for the oxide fuel;
- (b) however, for the metal fuel, the various contributions, and especially that of the fissile isotope, are increased by an appreciable amount. This may influence the Doppler effect if the fissile isotope is separated from the fertile one, as in some critical assemblies;
- (c) the fact that the contribution of the spatial shift of neutrons is negligible indicates that the heterogeneous Doppler effect can be computed as a local effect, i.e. as depending only on the flux and adjoint flux in the fuel region, provided that the diluent region is taken into account in the computation of the energy fine structure of the fuel region flux;
- (d) this last result justifies the approximation of ref. (1), since the spatial fine structure of the flux and adjoint flux is not so pronounced as to affect the Doppler effect appreciably.

Conclusions

A method was presented for the investigation of the effect of heterogeneity on the Doppler and sodium loss coefficients. Numerical results seem to show that heterogeneity has a negligible influence on the Doppler effect, although it may increase the individual contributions of the fissile and fertile isotopes. In contrast, heterogeneity seems to influence appreciably the sodium loss effect and to make it less positive.

3. A Simple Model for the Treatment of Neutron Slowing Down

Multi-group methods are not ideally suited for the description of neutron slowing down. On the one hand, owing to the complicated structure of the inelastic scattering operator, the physical features of inelastic slowing down are hidden under large amounts of numerical data. On the other hand, elastic scattering involves, in the absence of a moderator, mostly slowing down within the groups, which is not properly described by a multi-group approximation using the usual coarse groups.

Cadilhac and Pujol⁽⁹⁾ are developing a method which should give at the same time a simpler representation of the inelastic slowing down and a good representation of elastic slowing down, with a continuous energy variable.

In the slowing down operator $\Sigma_s(u')P(u' \rightarrow u)$, the transition probability $P(u' \rightarrow u)$ is replaced by $\tilde{P}(u' \rightarrow u)$ where

$$\tilde{P}(u' \rightarrow u) = \begin{cases} f(u') \cdot g(u) & \text{when } u \geq u' \\ 0 & \text{when } u < u'. \end{cases} \quad (1)$$

In order to conserve the neutron balance $\tilde{P}$ must satisfy

$$\int_{u'}^{\infty} \tilde{P}(u' \rightarrow u) du = f(u') \int_{u'}^{\infty} g(u) du = 1. \quad (2)$$

Introducing the auxiliary function $h(u) = 1/f(u)$, $\tilde{P}$ can be expressed as

$$\tilde{P}(u' \rightarrow u) = -h'(u)/h(u'). \quad (3)$$

In order to determine $h(u)$ an additional condition is used:

The slowing down source per unit lethargy is given by

$$\rho(u) = \int_0^u \Sigma_s P(u' \rightarrow u) \phi(u') du'. \quad (4)$$

$\tilde{P}(u' \rightarrow u)$ is chosen such that it has the same action as $P(u' \rightarrow u)$, i.e. it gives the same slowing down source $\rho(u)$, on a particular reference spectrum $\phi_0(u)$.

It is clear that if one chooses for $\phi_0(u)$ the asymptotic spectrum of a medium, the correct asymptotic slowing down source would be obtained with the approximate operator, even though the detailed neutron slowing down histories are not preserved.

For this method to be useful, however, the choice of $\tilde{P}$ should not depend on the *a priori* knowledge of the space dependent spectrum.

Cadilhac and Pujol have studied the sensitivity of some reactor parameters (k_{eff} , first and second moments of the generation time, the spectrum) to the reference spectrum $\phi_0(u)$, using the procedure described above only for the inelastic slowing down.⁽⁹⁾ They found the results to be practically insensitive to the choice of the reference spectrum. More work should be done on the elastic slowing down. If the total slowing down operator can be factorized, the integral slowing down equation can be replaced by a differential equation.⁽⁹⁾

4. The Application of Perturbation Theory to the Ratios of Fluxes and Reaction Rates^(10,11)

Introduction

In the study of large fast reactors and in the analysis of critical experiments it is frequently desired to obtain the variation of such ratios (e.g. the breeding ratio, $P_{\text{max}}/P_{\text{average}}$, spectral indices) for some variations of the reactor parameters. These parameters are of two types:

- the configuration or the composition of the reactor;
- the cross-sections and other input data.

As is well known, perturbation methods are of great interest when the effects of small changes in the parameters of a system are investigated. If

these small variations were to be obtained indeed by a difference between two direct calculations, a very great accuracy would be required on each calculation for their difference to be still meaningful. Perturbation methods yield a better accuracy in less time and give more insight into the physics of the phenomena and the relative influence of the different parameters involved.

Usachev⁽¹²⁾ has recently presented a perturbation method in which he investigates the change in one characteristic of a reactor (e.g. a reaction rate) for arbitrary changes in the reactor parameters (e.g. a composition). He defines an importance function of the neutrons for that characteristic, this importance being calculated as the sum of a convergent series. Then the variation of that characteristic is obtained through integration on the so-defined importance function.

Another method, somewhat dual to Usachev's one, is presented here, which investigates the change in any characteristic of a reactor, but for one change in the reactor parameters. The two methods are complementary, the choice between them depending on the relative numbers of output characteristics and of different reactor configurations to be studied. In this method the calculation of the flux perturbation constitutes an essential step. From the flux perturbation the variation of any reaction rate or other characteristic is easily obtained. The quantities of interest may be local (e.g. maximum power density) or integrated over a part or over the whole of the volume.

Description of the Method

Let a perturbation be introduced into a critical reactor by altering some independent parameter (concentration of a material for example) and let the reactor be kept critical by changing another dependent parameter (concentration of another material, enrichment, geometric buckling). The perturbation of some reaction rate is sought.

The critical reactor equation is

$$M\phi = 0. \quad (1)$$

For the reactivity change to be zero in first order, the perturbation dM must satisfy

$$(\phi^*, dM\phi) = 0 \quad (2)$$

where ϕ^* is the adjoint flux.

The variation of the dependent parameter is determined from this linear equation; dM is then completely defined.

A reaction rate R_K is written as

$$R_K = T_K \cdot \phi \quad (3)$$

where T_K is a linear functional, expressed in terms of cross-sections.

The first order perturbation of R_K is

$$dR_K = dT_K \cdot \phi + T_K \cdot d\phi \quad (4)$$

$d\phi$ is the only unknown quantity in this equation; it is evaluated from the criticality condition

$$d(M\phi) = dM \cdot \phi + M \cdot d\phi = 0 \quad (5)$$

or

$$M \cdot d\phi = -dM \cdot \phi. \quad (6)$$

This is the classical equation giving the first order flux perturbation (equation A5 of ref. (3)) particularized to a zero change of reactivity.

Equation (6) states that $d\phi$ is the stationary solution of the unperturbed reactor equation with a known external source $-dM\phi$. A stationary solution is possible for a critical reactor with source because this source has a zero importance, as shown by equation (2). This is a physical interpretation of equation (2) which is also, from the mathematical viewpoint, the condition for a non-trivial solution $d\phi$ of equation (6) to exist.⁽¹³⁾

The general solution of (6) is the sum of a particular solution $d\phi_{\text{part}}$ of the inhomogeneous equation (6) and an arbitrary multiple of the general solution of the homogeneous equation $M\phi = 0$

$$d\phi = d\phi_{\text{part}} + C \cdot \phi \quad (C = \text{constant}). \quad (7)$$

C could be determined by additional physical conditions (total power of the reactor, fission source at one point). However, $d\phi$ must be small because it was used in perturbation calculations.

When $d\phi$ is known, dR_K is immediately given by equation (4). For the ratio of two reaction rates R_1 and R_2 (e.g. the breeding ratio) the relative variation is, to first order

$$\frac{d(R_1/R_2)}{R_1/R_2} = \frac{dR_1}{R_1} - \frac{dR_2}{R_2} = \frac{T_1 \cdot d\phi_{\text{part}} + dT_1 \cdot \phi}{T_1 \cdot \phi} - \frac{T_2 \cdot d\phi_{\text{part}} + dT_2 \cdot \phi}{T_2 \cdot \phi} \quad (8)$$

where the constant C dropped out. Thus it appears that one can choose for $d\phi$ any particular solution as long as it is small.

Application of the Method

The method described above is quite general. It will be applied to space dependent problems, by the appropriate modifications of the usual multi-group codes with external sources. In a first step it has been applied to fundamental mode calculations with the multi-group representation. The machine code PERAS has been developed which calculates the flux perturbation and the variation of arbitrary reaction rates, or of their ratios.

When solving equation (6) the code makes equal to zero the component

of $d\phi$ for the energy group in which the non-perturbed flux is maximum. It has been found that this procedure yields the best accuracy in the calculations and guarantees that $d\phi$ remains small.

Some numerical results are presented here. They refer to a fast oxide dilute reactor, fuelled with Pu-239 and U-238 (enrichment 12.5%); the reactor is perturbed by the homogeneous insertion of U-235, corresponding to a change in the depleted uranium composition. The reactor is held critical by varying the buckling Bm^2 accordingly.

In order to check the limits of applicability of perturbation theory, five cases have been considered, corresponding to the addition of U-235 with the following atomic densities (in units of 10^{24}): 5×10^{-6} , 10^{-5} , 5×10^{-5} , 10^{-4} , 5×10^{-4} (U-238: 0.74×10^{-2}). Calculations were made with the Hansen and Roach cross-section set. For the cases $n_{u5} = 10^{-5}$ and $n_{u5} = 10^{-4}$, for instance, the direct calculation gives a change of the buckling with respect to the unperturbed case of 1.929% and 18.757% respectively, while PERAS gives 1.935% and 19.35%.

For the group fluxes the results obtained for the 9 first groups only are quoted since there are practically no neutrons in the other groups. For $n_{u5} = 10^{-5}$ and 10^{-4} the relative group flux variations attain 2% and 20%, respectively, and the relative errors of the PERAS results do not exceed 0.03% and 3%, respectively.

The effect on the breeding ratio BR has been calculated for $n_{u5} = 10^{-5}$

$$BR = \frac{\langle \Sigma_{C_{u5}}, \phi \rangle}{\langle (\Sigma_{C_{Pu9}} + \Sigma_{f_{Pu9}}), \phi \rangle}$$

It is found

$$\left(\frac{dBR}{BR} \right)_{\text{part}} = (-0.4226 \times 10^{-2}) - (-0.2721 \times 10^{-2}) = -0.1505 \times 10^{-2}$$

where the first and second terms refer respectively to the capture in U-238 and the absorption in Pu-239; while a direct calculation yields

$$\frac{d(BR)}{BR} = -0.150 \times 10^{-2}.$$

5. Control Rod Calculations

The straightforward use of diffusion theory for calculating control rods worths in fast reactors has the disadvantage of introducing an unknown error while more accurate transport calculations in one or two dimensions are very time consuming.

Lacapelle⁽¹⁴⁾ has developed a method based on diffusion theory but with a modified diffusion coefficient, which is an extension to multi-group of the

one-velocity ABH method⁽¹⁵⁾ used for the determination of the thermal utilization factor f . In this method the group diffusion coefficients of the rod are adjusted so as to obtain with diffusion theory the correct average fluxes and hence the correct number of captures in the rod.

A numerical example is presented here in order to illustrate the accuracy of this method. A control rod is placed at the centre of a cylindrical cell ($r = 106.25$ cm) with the composition of the central zone of the 1000 G reactor ($k_{\infty} = 1.028$).⁽¹⁶⁾ Six cases were considered and the results are presented in Table 3. It is seen that while ordinary diffusion theory gives discrepancies of up to 9%, the method presented here agrees with S_{16} calculations within 1%.

TABLE 3. RELATIVE DIFFERENCES BETWEEN THE CONTROL ROD WORTHS GIVEN BY THIS METHOD TAKEN AS REFERENCES AND BY OTHER METHODS

	Radius	Ordinary diffusion theory (%)	DSN	
			S8 (%)	S16 (%)
Natural boron carbide	3.125	+4.5	-0.01	-0.03
	6.25	+4.3	+0.5	+0.4
	12.5	+3.7	+1	+0.8
Enriched boron carbide	3.125	+9.2	+1	-0.1
	6.25	+5.7	-0.4	-0.5
	12.5	+3.6	-0.7	

6. One Group Calculations

Parametric studies and long term behaviour analysis require long computer times when done in multi-group. Since for large fast reactors the spectrum does not vary much throughout a given region, and is not very sensitive to small variations around a reference composition and to the burnup, some calculations can be performed using a one group model with large savings in computer time and in preparation and interpretation work. A one group one dimensional analytic code UNDIGO⁽¹⁷⁾ was developed in Cadarache for this purpose. The one group constants can be obtained in one of three ways:

- weighting by an average region-spectrum obtained from a reference multi-group calculation;
- weighting by the asymptotic region-spectrum;
- interpolation from one group tables for various compositions.

In this code some specific subroutines are incorporated for performing such studies as power flattening by zoned loadings and long time behaviour

of such loadings.⁽¹⁸⁾ A remarkable agreement was found, particularly in the case of a 1000 MW(e) reactor, and somewhat less in the case of a 250 MW(e) reactor, between one group and multi-group calculations. Optimization studies performed with the one group code also yielded almost exactly the desired optimum values.

A numerical version of the one group code should prove very appropriate for the treatment of three dimensional problems.

7. Multi-group Cross-sections Set

A 25-group cross-sections set is at present being prepared at Cadarache.

Basic data used come from evaluation work done in part at Cadarache in collaboration with Electricité de France and in part in other countries, in particular at Winfrith and Aldermaston.

Ni⁽¹⁹⁾ and Cr⁽²⁰⁾ have been exhaustively studied at Cadarache and a similar study for U-238 is under way. These evaluations are incorporated in the Winfrith Nuclear Data File. Group cross-sections⁽²¹⁾ are derived from the basic data through a weighting on a flux having a structure $1/\Sigma_t(E)$ multiplied by a slowly varying function of E . This procedure is applied as well to the resonance absorption as a function of temperature as to the slowing down phenomena.

After its completion this set will be tested by calculating critical experiments (ZPR-III, ZEBRA, BFS).

8. Low Energy Tail of the Fission Spectrum

As mentioned in the introduction, calculations always underestimate the low energy tail of the flux. We are investigating whether part of the error might be due to the assumed fission spectrum. Because of the absence of a moderator and because of the large fuel concentration the resonance absorption is so large in a fast reactor that the low energy spectrum is very sensitive to the corresponding part of the fission spectrum. In most cross-sections sets, the fraction of fission neutrons born below 100 keV is about 1.4%. Leroy and Quentin⁽²²⁾ are investigating whether this fraction might be higher. Their work includes a careful review of past experimental and theoretical work, and a detailed calculation in the laboratory system of the energy of the neutrons which evaporate from the moving fission fragments in the backward direction.

9. Monte Carlo Code for Criticality

Monte Carlo methods are ideally suited for solving problems with complex geometries, for instance those encountered in the criticality safety area.⁽²³⁻²⁵⁾

The statistical precision obtainable with present large computers allows the use of these methods in fast reactor physics calculations. They are particularly suited to the determination of the critical mass in the exact geometric configuration, including the lattice structure.

A tridimensional code, with meshes in the x - y plane, has been developed; it treats up to 30,000 rectangular meshes of at most sixty-four different types; each mesh can be cut by sixty-four planes parallel to the x - y plane and can also contain cylinders with axis parallel to o_z . Cross-sections may be taken from a multi-group set or from a continuous energy library.

Typical running times for a standard deviation of $\Delta k/k = 2 \times 10^{-3}$ are about 1 hour on an IBM 7094, approximately the same time as a two dimensional transport code (TDC). Running times are practically independent of the geometry and of the number of groups.

If a continuous energy library with a resolution good enough to describe scattering resonances could be used, the Monte Carlo method would be the only one to treat correctly the effect of these resonances near the interfaces.

10. Doppler Effect Calculations

Overlap Effect

Neviere and Ravier have investigated this effect.^(26,27) The Doppler effect around 1 keV was determined with the N.R. approximation with overlap taken into account, including the flux correction factor.

The resonance parameters which were used in the calculation were not derived from statistical laws, but were for U-238, those actually measured around 1 keV⁽²⁸⁾ and for Pu-239, those measured at lower energies⁽²⁹⁻³¹⁾ shifted upwards to 1 keV. Furthermore, the effective cross-sections were calculated by numerical integration, using Doppler broadened Breit-Wigner single level formulae with potential scattering phase shift taken into account.

It was found that the overlap has practically no effect on the negative U-238 Doppler effect, but that it reduces appreciably the positive Pu-239 Doppler effect, which may even become negative. The negative capture component of the Pu-239 Doppler effect is slightly reduced while the fission component is strongly affected, and may by itself become negative. This seems to be mainly due to the fact that the Pu resonances belong to two families of which one has a large average fission width (730 MeV).⁽³²⁾ These wide resonances are not very sensitive to Doppler broadening, but are particularly sensitive to the overlap effect. The calculations show that the net Doppler coefficient becomes more negative by the overlap effect. For instance, for the 1000 G reactor,⁽¹⁶⁾ the Doppler effect is more negative by 5% between 300°K and 900°K, and by 10% between 900°K and 3000°K.

Moments Method

A new method is being investigated by Cadilhac and Nevière to calculate the effective cross-sections

$$\tilde{\mu}_x = \frac{\langle \mu_x / \mu_T \rangle}{\langle 1 / \mu_T \rangle}$$

which enter in the calculation of the Doppler effect.

Since it is easier to evaluate moments of positive order: $m_1 = \langle \mu \rangle, \dots, m_n = \langle \mu^n \rangle$ an algorithm has been found to calculate $m_{-1} = \langle 1/\mu \rangle$ as a function of $m_1, \dots, m_n$ thus yielding a method which is valid as well for isolated and overlapping resonances. The distribution function F is calculated for known resonances and the various moments $m_n = \int \mu^n dF$ are calculated as a function of temperature. The effectiveness of this method has been investigated and has been found satisfactory even for U-238 at low energy with the first five moments.

Theory of the Measurement

The more theoretical aspects of the measurement of the Doppler effect were treated in ref. (33). Khairallah⁽³⁴⁾ has investigated the accuracy which can be obtained in the Doppler coefficient measurement when a hot and cold sample are interchanged in a critical assembly.

11. Analysis of Multi-zone Experiments

Difficulties are encountered in the interpretation of zoned experiments particularly for the determination of the material buckling by flux mapping, when the experimental zone is so small that flux transients propagate from the interface over a large fraction of the experimental zone. A progressive substitution method has proved very useful to increase the accuracy in the case of thermal neutron systems.⁽³⁵⁻³⁷⁾ The extension of this method to fast systems is being investigated.⁽³⁸⁾ Progressive substitution experiments give a curve, rather than a point, and should therefore give a better understanding of the effect of the interface. Furthermore, this effect can be to some extent eliminated by this method when the asymptotic characteristics of the experimental zone are sought.

The interface flux transients can be divided into three types:

- coarse group transients, which can be treated by diffusion or transport multi-group codes;
- transients within the coarse groups, caused particularly by the scattering resonances;
- narrow energy transients corresponding to the narrow absorption resonances of the heavy nuclei.

The second and third type transients cannot be properly treated at this time, particularly the second type transients which are not erased by one collision.

It is clear that their theoretical treatment and the interpretation of progressive substitution experiments will benefit from one another.

12. Pulsed Experiments

A large amount of theoretical work has been carried out on the analysis and the interpretation of pulsed experiments on fast systems. The conditions of existence of an asymptotic mode and the possibility of measuring pseudo-fundamental modes have been discussed in refs. (39) and (40); the physical meaning of the distinction between those modes is summarized in ref. (41). This work, illustrated first in multi-group, has been extended with the continuous energy model by Cadilhac and Nicolaenko.⁽⁴²⁾

The method developed for the study of pulsed experiments is quite general for fast systems⁽⁴³⁾ and was used also for the study of different types of modulated experiments.⁽⁴⁴⁾ Computer codes have been written in FORTRAN for the comprehensive analysis of these experiments.⁽⁴⁵⁾

13. Safety and Stability

Work is continuing on the study of the events that could lead to the fusion of the core, on the calculation of the corresponding reactivity insertion rates, and on the evaluation of the resulting accident. The influence of the Doppler coefficient on the first stage of the accident and of the equation of state is being investigated. A code has been developed⁽⁴⁶⁾ in which special care is taken in giving a correct description of the thermodynamics. In particular the latent vaporization heat is taken as a function of temperature and the space dependent temperatures as well as the pressures are given at each time step, thus allowing the calculation of both feedback components.

The code treating the stability with a point neutronics model and a space dependent feedback is now in routine operation.⁽⁴⁷⁾ Some work was done with a spatial neutronics model in order to investigate the possibility of spatial instabilities and of flux tilt.⁽⁴⁸⁾ With the simplified model used, it appeared that spatial instability could not occur in currently designed fast reactors.

14. Shielding (contribution of M. RASTOIN)

Two main problems have to be considered. First, the attenuation of gamma and neutron fluxes to protect structural materials so that they keep their characteristics during all the reactor life time^(49, 50) and to design cooling circuits for the shields.

The second problem is the determination of the biological shield for zones where access may be required when the reactor is operating or shut down, and the determination of the activity of coolants containing radioactive nuclides, and especially the evaluation of the repartition of fission and corrosion products in circuits.

Reactor Design—Importance of the Choice of Shielding Materials

In the conceptual design stage, the shield is an important factor in the selection of a solution, since it assigns qualitatively the nature of materials as well as the geometry suitable for each reactor concept.

For a fast reactor, shielding materials must have good slowing down properties and be weak emitters of capture gamma rays: e.g. graphite, borated graphite, boron carbide, boron steel, concrete. Concretes are especially studied from the point of view of their water content as a function of the temperature.

The shield of Rapsodie is made of concrete containing serpentine, corundum and rare earths, the operating temperature is 250°C. The cost per kilogram of shielding materials varies considerably, from 200 francs (\$40) a kilo for boron carbide to a fraction of a franc for concretes and sands. It is important to define exactly the admissible dose rates for the different zones during reactor operation and after shutdown.

The geometry of the shield may differ considerably in function of the solution chosen for the cooling sodium circuit. Two different solutions are studied:

- attenuation of the radiations by incorporating the coolant pipe in a series of attenuating cavities;
- or by flowing the coolant through an heterogeneous lattice of cylinders containing a shielding material.

Another very important heterogeneity of the shield arises from the openings necessary for fuel handling and movement of control rods.

Shielding studies can therefore be divided into two groups:

- attenuation studies in materials;
- penetration studies in gaps and cavities.

Attenuation Studies in Homogeneous Materials

For fast reactors, neutron slowing down phenomena and absorption at intermediate energies require to be considered with special care.

This implies the use of transport codes (NIOBE in spherical geometry^(51, 52) and Monte Carlo codes (in spherical geometry SANE⁽⁵³⁾ and SIFER; in cylindrical geometry SIDEF).

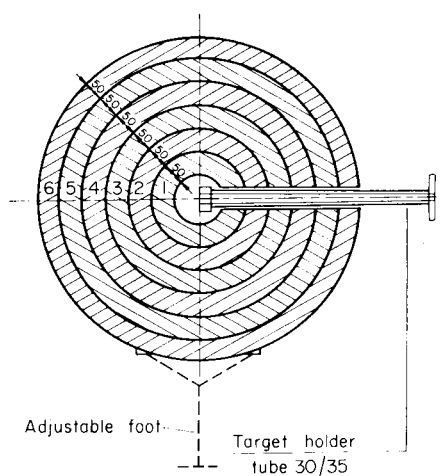


FIG. 1

Silicon *n.p.* equivalent fission flux
Intensity source = 10^{10} n/s

$$\frac{\Sigma 14 \text{ Mev}}{\Sigma f} = 40$$

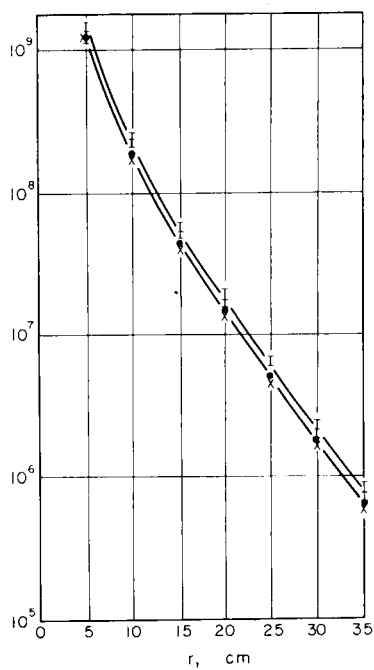


FIG. 2

The actually available compilations of cross-sections will probably soon be satisfactory, but further work is required to store them in the computer and to select them when needed. The larger fast memory of the new generation computers as well as the reduction of the access time will help to solve this problem.

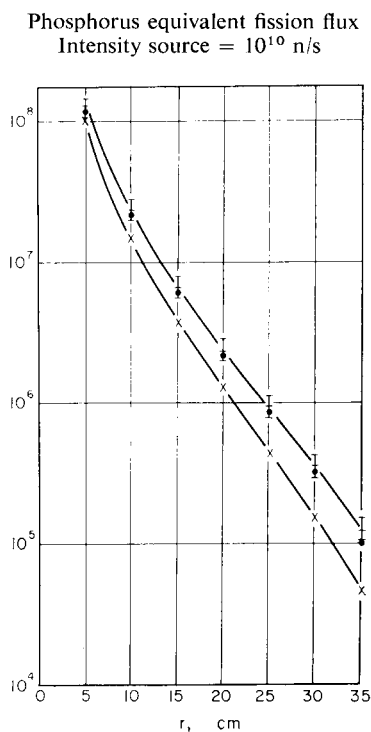


FIG. 3

Iron and cast iron are most attractive shielding materials for gamma radiation, but although they are good attenuators for fast (>1 MeV) as well as thermal neutrons, they are less effective for intermediate neutrons.

Recently computer codes using the Monte Carlo method to calculate neutron attenuation have been written in the United States (SANE) as well as in France (SIFER). It is difficult to evaluate the validity of these codes without a comparison between the results of calculations and a special experiment. At Fontenay-aux-Roses a SAMES accelerator was used, emitting 14 MeV neutrons in a spherical iron block (Fig. 1). In this case the simple geometry allows the comparison of different cross-sections sets and a test of the various approximations being used.⁽⁵⁴⁾

Silicon equivalent fission flux

The two codes gave the same results as the experiment. One may conclude

that the cross-sections for iron are correct at 14 MeV, since the fast flux propagation is well predicted (Fig. 2).

Phosphorus equivalent fission flux

There is agreement between experiment and results obtained with SANE. The flux calculated by SIFER is systematically lower, the discrepancy growing with the distance (25% at 5 cm, 60% at 35 cm) (Fig. 3).

One may attribute this discrepancy to the energy spectrum of the neutron after inelastic and $(n, 2n)$ collisions, the maximum of which for SIFER is below the threshold of the phosphorus detector, while SANE gives a higher value for the mean neutron energy (Fig. 4).

Rhodium equivalent fission flux

The code SIFER and the experiment give the same result. The flux given by SANE is lower than the experimental flux by 25–30% (Fig. 5).

Rhodium equivalent fission flux
Intensity source = 10^{10} n/s

$$\frac{\Sigma 14 \text{ Mev}}{\Sigma f} = 0.6$$

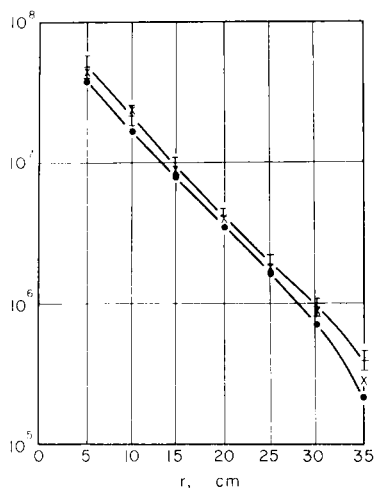


FIG. 4

This difference is due especially to the fact that one of the neutrons produced by the $(n, 2n)$ reaction is not taken into account in the SANE code.

Conclusion

The simplicity of this experiment from the geometrical point of view, as well as from the point of view of the neutron source used, allowed it to be

reproduced completely by a Monte Carlo calculation without any simplifying assumption.

A comparison in absolute value between the experiment and the calculation could be realized.

In general, taking into account the new detector coefficients measured at 14 MeV, the calculations are in good agreement with the experiment.

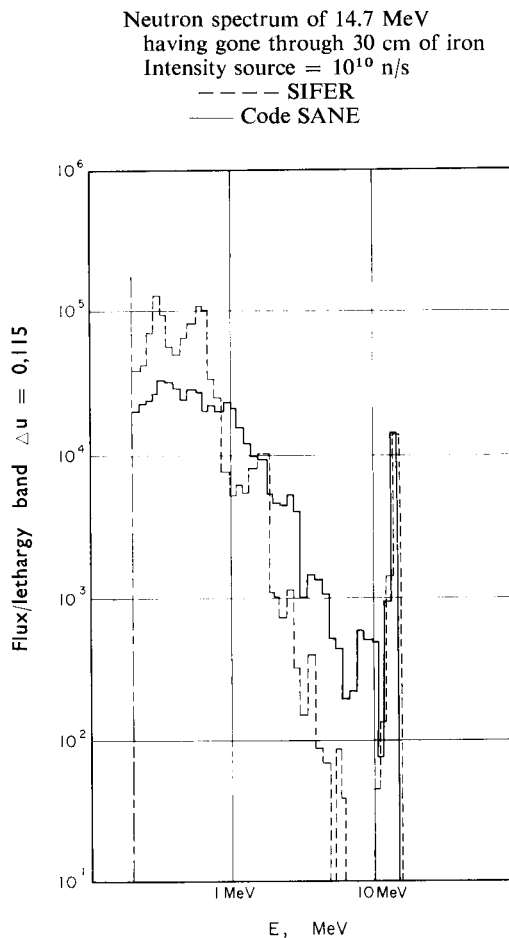


FIG. 5

The slowing down of monoenergetic neutrons of 3.5 and 14.7 MeV in a sodium block of dimensions $150 \times 150 \times 250$ has also been studied.⁽⁵⁵⁾ The calculations are made with the code SANE (Figs. 6 and 7).

The attenuation of neutrons in concretes heated up to 450°C was studied by means of the Naïade I facility.

Slowing down of neutrons of 14.7 MeV in sodium

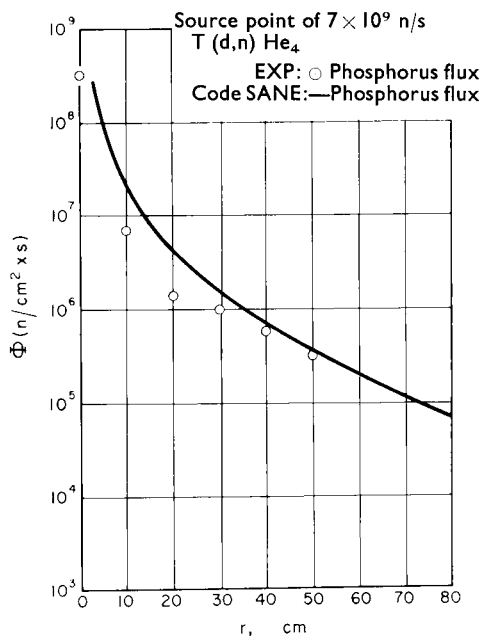


FIG. 6

Slowing down of neutrons of 3.5 MeV in Sodium

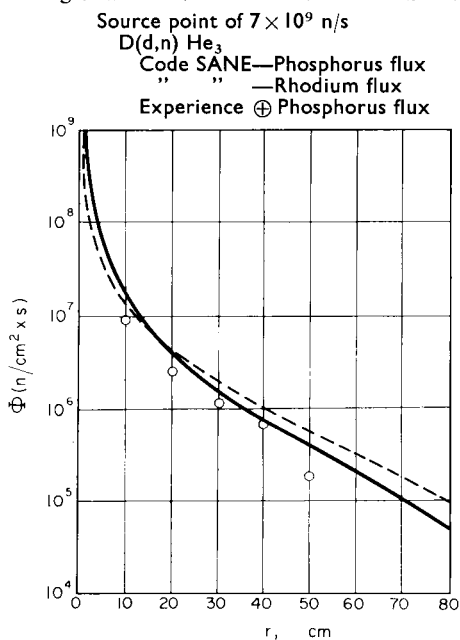


FIG. 7

Experimental studies on the attenuation of radiations in heterogeneous zones with sodium will be realized in 1967 with Harmonie at Cadarache.

Study of the Propagation in Gaps and Cavities

From the theoretical point of view two methods are now being considered:

a three dimensional Monte Carlo calculation using the code GMCM 9 for cavities;⁽⁵⁶⁾

a method using a Monte Carlo calculation to solve the geometric problem, the wall of the gap being represented by a reflecting surface, the reflection coefficients being calculated elsewhere.^(57,58)

References

1. P. GREEBLER and E. GOLDMAN, *Doppler Calculations for Large Fast Ceramic Reactors* GEAP 4092 (Dec. 1962).
2. D. MENEGHETTI, Calculation studies of effects of thin-slabs heterogeneities upon spectral reactivity component of Na-voiding, *Trans. Am. Nucl. Soc.* **8**, 2 (1965).
3. F. STORRER, A. KHAIRALLAH, M. CADILHAC and P. BENOIST, Heterogeneity calculation for fast reactors by a perturbation method, *Nucl. Sc. Eng.*, **24**, 2, 153-65 (Feb. 1966).
4. A. KHAIRALLAH and F. STORRER, to be published.
5. L. P. ABAGYAN *et al.*, *Group Constants for Nuclear Reactor Calculations*, Consultant Bureau, New York (1964).
6. T. LACAPELLE, internal CEA report (June 1964).
7. P. BENOIST, *Théorie du coefficient de diffusion dans un réseau comportant des cavités*, CEA report No. 2278 (Jan. 1964).
8. E. A. FISCHER, *Calculation of the Doppler Coefficient for Pu-oxide fueled Fast Reactors*, APDA 157 (1963).
9. M. CADILHAC and M. PUJOL, A simple model for the inelastic scattering of fast neutrons, to be published.
10. F. STORRER, *Méthode de perturbation pour le taux de régénération et autres rapports de taux de réaction*, Report ASS.EUR/CEA PNR/SECNR 65.01 (Feb. 1965).
11. F. STORRER, F. EBERSOLDT and P. GOVAERTS, to be published.
12. L. N. USACHEV, *Atomnaya Energiya*, Dec. 1963, pp. 472-81.
13. B. FRIEDMAN, *Principles and Techniques of Applied Mathematics*, Chapter I, J. Wiley & Sons, New York (1956).
14. T. LACAPELLE, to be published.
15. A. AMOUYAL, P. BENOIST and J. HOROWITZ, Nouvelle méthode de détermination du facteur d'utilisation thermique d'une cellule, *Journal Nucl. Energy* **6**, 79-98 (1957).
16. P. P. CLAUZON and C. GIACOMETTI, *Etudes neutroniques de cœurs de réacteurs rapides de 1000 MW(e) à deux zones d'enrichissement différent. Description du cœur de référence "1000 G"*, Report ASS. EUR/CEA PNR/SEPR 65.028 (Sept. 1965).
17. C. GIACOMETTI and P. P. CLAUZON, "UNDIGO" code de calcul à un groupe et une dimension en théorie de diffusion, Report ASS.EUR/CEA PNR/SETR, to be published.
18. C. GIACOMETTI, P. P. CLAUZON, G. GOURIEVIDIS and G. KESSLER, *Etude de la composition isotopique du Pu à l'équilibre dans les réacteurs de la filière "neutrons rapides"*, Report ASS.EUR/CEA PNR/SEPR 65.040 (Nov. 1965).
19. J. RAVIER and M. VASTEL, *Sections efficaces neutroniques du Nickel dans le domaine d'énergie 0,000 1 eV à 15 MeV*, Report ASS.EUR/CEA PNR/SEPR 65.010 (Mar. 1965).
20. J. RAVIER and M. VASTEL, *Sections efficaces neutroniques du Chrome dans le domaine d'énergie 0,000 1 eV à MeV*, Report ASS.EUR/CEA PNR/SEPR 65.041 (Nov. 1965).
21. J. RAVIER, *Sections efficaces multigroupes et jeu SEPR à 25 groupes*, Report ASS. EUR/CEA PNR/SEPR 66.050 (Feb. 1966).

22. J. L. LEROY and QUENTIN, Report SAMNF Cadarache, to be published.
23. M. F. ITHURRALDE, J. KREMSEY, J. LECLERC, C. LOMBARD, J. MOREAU and C. ROBIN, *Interprétation des expériences de criticité sur des solutions homogènes de plutonium et d'uranium*, Report CEA R 2488.
24. J. MOREAU, H. RABOT and C. ROBIN, *Méthode de Monte Carlo. Codes pour l'étude des problèmes de criticité*, Report CEA R 2872.
25. J. MOREAU, B. PARISOT, H. RABOT and C. ROBIN, Evolution des méthodes de calcul en criticité, méthode de Monte Carlo, interaction neutronique, *Colloque AIEA sur la Prévention de la Criticité des Produits Fissiles*, Stockholm (Nov. 1965).
26. G. NEVIÈRE, J. RAVIER and J. M. CHAUMONT, Effect of resonance overlap on the Doppler coefficient for large fast reactors, to be published.
27. G. NEVIÈRE and J. RAVIER, *Effet d'interférence des résonances de l'U238 et du Pu239 sur les sections efficaces effectives et le coefficient Doppler d'un réacteur à neutrons rapides*, Report ASS.EUR/CEA PNR/SETR R 002 (May 1966).
28. J. B. GARG, J. RAINWATER, J. S. PETERSEN and W. W. HAVENS, JR., Neutron resonance spectroscopy: Th 232 and U238, Columbia University, *The Physical Review*, June 8, 1964.
29. G. DE SAUSSURE, J. BLONS, C. JOUSSEAUME, A. MICHAUDON and Y. PRANAL, Mesure de la section efficace de fission du Plutonium 239 de 0 à 5 KeV, *Colloque sur la Physique et la Chimie de la Fission SM/6013, Salzburg, Austria, May 1965*.
30. J. BLONS, H. DERRIEN, A. MICHAUDON, P. RIBON and G. DE SAUSSURE, Study of resonance parameters of Pu 239, *International Conference on the Study of Nuclear Structure with Neutrons, Antwerp, July 1965*.
31. C. A. UTTLEY, Single level analysis of the resonances in the total cross sections of Pu 239, *ibid.*
32. J. BLONS, H. DERRIEN, A. MICHAUDON, P. RIBON and G. DE SAUSSURE, *Comptes-rendus de l'Académie des Sciences* **262**, No. 1 (Jan. 1966).
33. F. STORRER, A. KHAIRALLAH and J. OZEROFF, *Theoretical Aspects of the Measurement of the Doppler Coefficient in a Critical Facility*, ANL 6792 (Dec. 1963).
34. A. KHAIRALLAH, Evaluation de la précision de la mesure de l'effet Doppler dans Masurca, PNR/SETR internal memo (Sept. 1965).
35. P. BACHER and R. NAUDET, Mesures de Laplaciens par la méthode du remplacement progressif, *J. Nucl. Energy, Part A*, **13**, 112-27 (1961).
36. P. BACHER, *Remarques sur l'interprétation des expériences de remplacement progressif dans MARIUS et sur la comparaison avec les expériences PCTR-CEA Service de Physique Mathématique*, Report SPM No. 702 (Nov. 1961).
37. R. NAUDET, *Examen critique des méthodes de substitution*, EACRP-L-32.
38. M. BARBERGER, P. CAUMETTE, P. GOVAERTS and F. STORRER, Méthodes de mesure et utilisation du Laplacien matière dans les milieux à neutrons rapides, to be published.
39. F. STORRER and M. STIEVENART, *Contribution to the Theory of the Pulsed Neutrons Technique Applied to Fast Multiplying Systems*, Report EUR/C/593 e.
40. F. STORRER, Pulsed neutron experiments on fast and intermediate systems, SM 62/88, *IAEA Meeting on Pulsed Neutrons, Karlsruhe, May 1965*.
41. F. STORRER, P. GOVAERTS and P. HAMMER, report to be published.
42. M. CADILHAC, P. GOVAERTS, P. HAMMER, M. MOORE, B. NICOLAENKO and F. STORRER, Contribution à la théorie des expériences pulsées et modulées sur des milieux multiplicateurs à neutrons rapides, SM 62/86, *IAEA Meeting on Pulsed Neutrons, Karlsruhe, 1965*.
43. F. STORRER, P. GOVAERTS, F. EBERSOLDT and P. HAMMER, A method for calculating the flux evolution in a fast neutron multiplying medium as a function of time, space, and energy, *Nucl. Sc. Eng.* **24**, 4, 344-8 (Apr. 1966).
44. M. CADILHAC, P. GOVAERTS, P. HAMMER, M. MOORE, B. NICOLAENKO and F. STORRER, *Etude comparative de plusieurs types d'expériences pulsées et modulées sur des milieux multiplicateurs à neutrons rapides*, SM 62/87.
45. P. GOVAERTS, P. HAMMER and J. HERAIL, *Méthodes de calcul des expériences pulsées dans l'approximation multigroupe*, Report ASS.EUR/CEA PNR/SETR R.001 (Apr. 1966).

46. I. PUIG, Code permettant le calcul des accidents de type explosif, ASS.EUR/CEA internal memo (Apr. 1966).
47. I. PUIG, Stabilité et fonctions de transfert des réacteurs rapides, ASS.EUR/CEA internal memo SEPR 63/2128.
48. S. AN, Note on spatial instability of large fast reactors, ASS.EUR/CEA internal memo (Oct. 1965).
49. P. DULIEU and J. RASTOIN, *Influence de la forme du spectre de neutrons sur l'effet Wigner dans les protections*, Note CEA No. 422.
50. P. DULIEU and J. RASTOIN, Influence du spectre et du flux intégré de neutrons sur les propriétés mécaniques de l'acier (modèle de Brinkman—Seeger), Note CEA à paraître.
51. *Advanced Shield Computational Techniques*, NIOBE UNUCOR 631-2.
52. G. BRANDICOURT and J. CULAMBOURG, Programme NIOBE, Note CEA No. 583.
53. *Advanced Shield Computational Techniques*, SANE UNUCOR 633-4.
54. D. CHOL, Atténuation des neutrons dans un massif de fer, Report CEA à paraître.
55. M. LOTT, A. TATTEGRAIN and H. SATGE, Ralentissement des neutrons dans un massif de sodium, Note CEA à paraître.
56. M. KNIEDLER and T. JORDAN, *Generalized Monte Carlo program for neutrons GMCM 9é*, MND-MC-2856.
57. G. HASSELIN, *Contribution à l'étude de la propagation des neutrons dans des cavités*, Report CEA R.2774.
58. C. DEVILLERS, ZEUS. Code de Monte Carlo à trois dimensions, Thèse à paraître.